



**Dealing with Uncertainty:
Enhancing Investment Decision-
Making and Resilience in Complex
Systems**

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Understanding Uncertainty and Human Factors

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- Balancing human judgment with analytical tools



Turning Insight into Value

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Uncertainty and Decision Making



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Investment Errors as a System of Systems Problem



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Table 4: The Olympic Games have the largest cost overrun of any type of large-scale project, real terms

	Roads	Bridges, tunnels	Energy	Rail	Dams	IT	Olympics
Cost overrun	20%	34%	36%	45%	90%	107%	172%
Frequency of cost overrun	9 of 10	9 of 10	6 of 10	9 of 10	7 of 10	5 of 10	10 of 10
Schedule overrun	38%	23%	38%	45%	44%	37%	0%
Schedule length, years	5.5	8.0	5.3	7.8	8.2	3.3	7.0

Source;
Regression to the Tail: Why the Olympics Blow Up, by Flyvberg, Budzier and Lunn September 2020

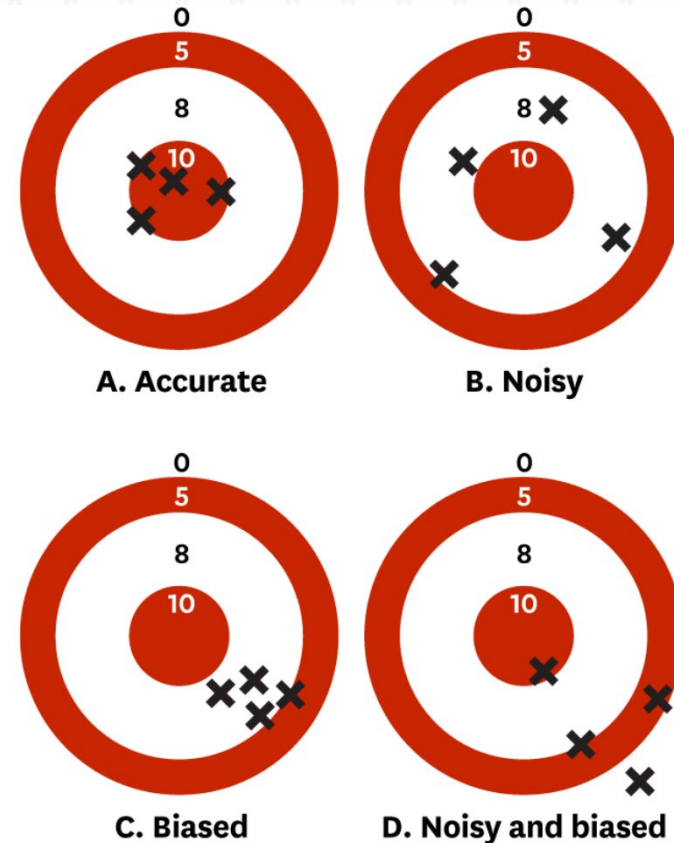
Why Human Factors Matter in Investment Decisions



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Projects fail more often from human errors such as **bias** and **noise** more often than from technical flaws.

Understanding psychology helps us reduce errors and make better investment decisions.



Individual Decision-Making

The Role of the Nervous System



Ensure the perpetuation of the species

To achieve that, it does two things:



1. Firstly, it protects from threats



2. Secondly, it seeks reward

Balancing both responses builds resilience under uncertainty.

Key Brain Regions for Decision-Making



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Key Brain Regions in Decision-Making

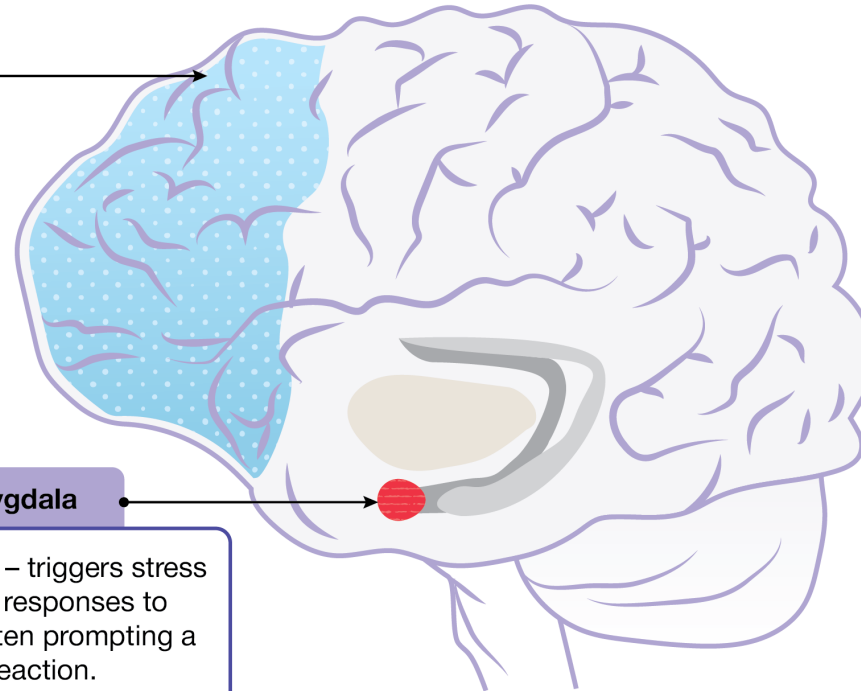


Prefrontal Cortex

The rational thinker – manages logical decision-making, planning, and impulse control. It helps balance emotions with reason during uncertainty.

Amygdala

The fear alarm – triggers stress and emotional responses to uncertainty, often prompting a fight-or-flight reaction.



Decision Making Under Uncertainty

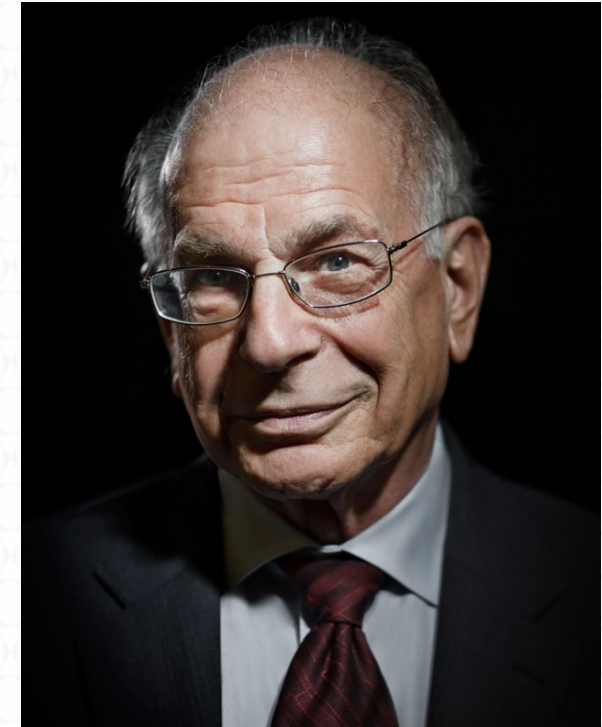


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There are two ways
we make choices:

Fast, intuitive,
automatic thinking
(System 1)

Slow, rational,
calculating thinking
(System 2)



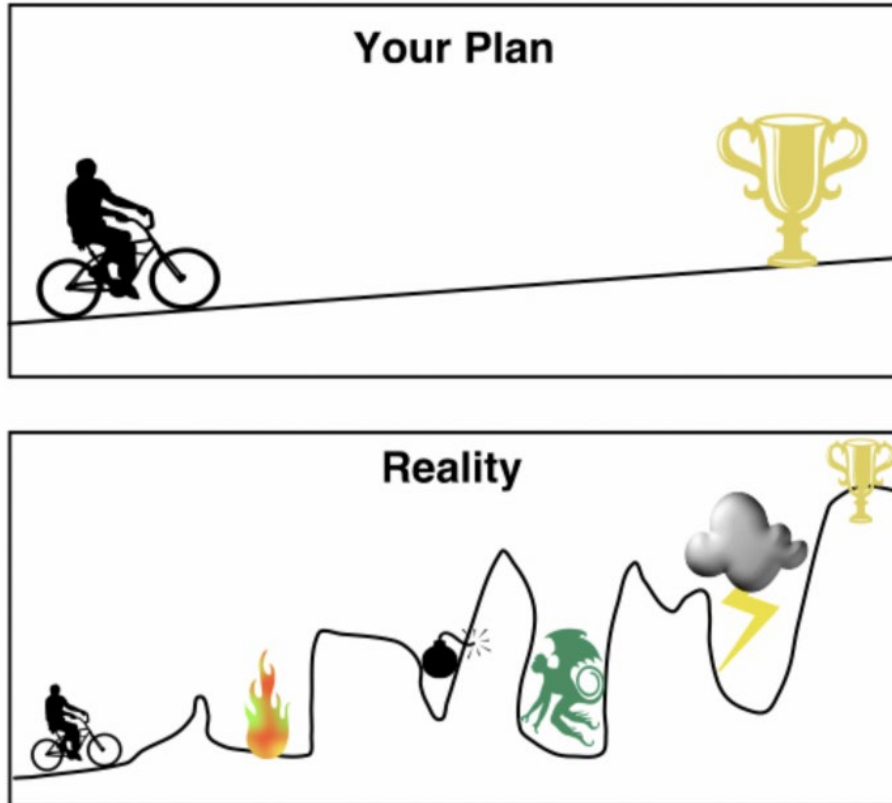
Our minds are flawed by errors and bias. This is particularly evident when dealing with complex problems.

Unconscious Bias

e.g. The Planning Fallacy



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The Planning Fallacy is a phenomenon in which predictions about how much time and costs will be needed to complete a future project display an optimism bias and underestimate the time and costs needed, and overestimate the benefits. (Kahneman, Tversky, Lovallo)

(source: [Wikimedia](#))

Unconscious Bias

e.g. Loss Aversion



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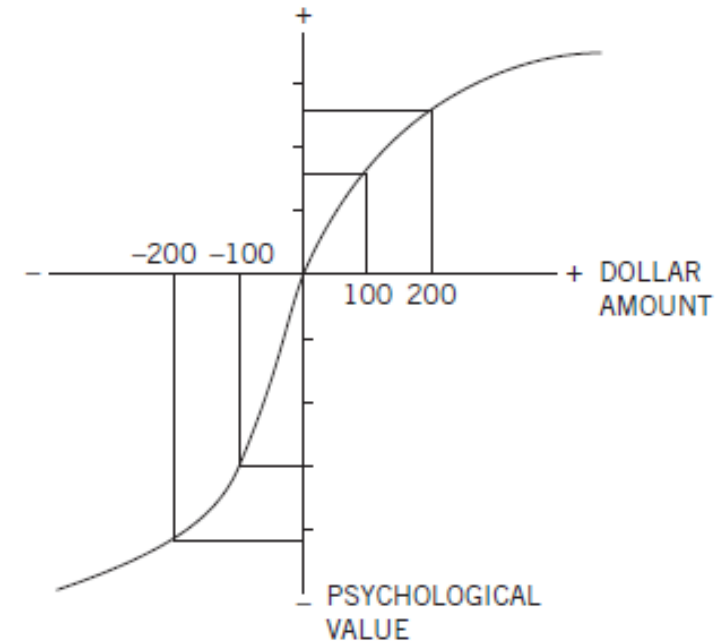
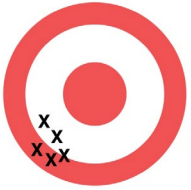


Fig. 10

The psychological value of gains and losses, which are the “carriers” of value in Prospect Theory. The response to losses is stronger than the response to corresponding gains. This is loss aversion.

How to mitigate common biases*



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 Bias	 Definition	 Example	 Mitigation
Confirmation Bias	Favouring information that fits beliefs	Ignoring critical investment reports	Seek opposing views and include diverse perspectives
Optimism Bias	Overestimating positive outcomes	Assuming a project will finish on time and within budget	Test assumptions with data
Availability Heuristic	Judging risk by what comes to mind	Fearing flying after a crash is on the news	Check statistics; don't rely on memory alone
Loss Aversion	Fearing losses more than gains	Sticking to old tools despite cost	Weigh long-term benefits
Anchoring Bias	Clinging to initial estimates	Fixating on early project costs	Reassess with fresh data

Think of a time when your first impression or gut instinct turned out to be wrong.

Keys to reducing project errors and developing expert intuition



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Regular world

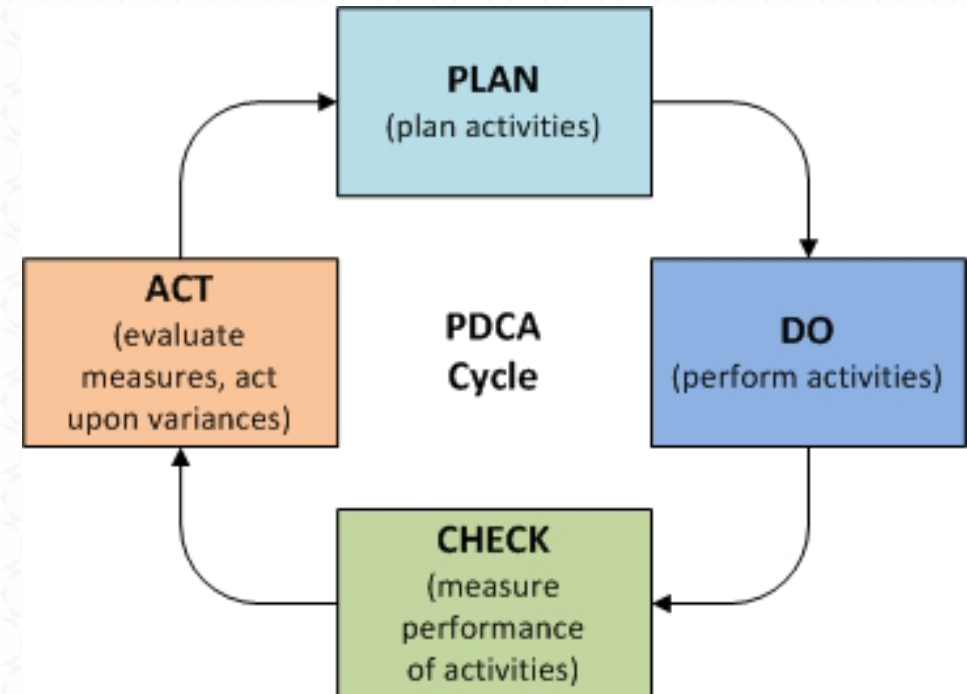
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Many opportunities to learn

+

Immediate Feedback

Expert Intuition



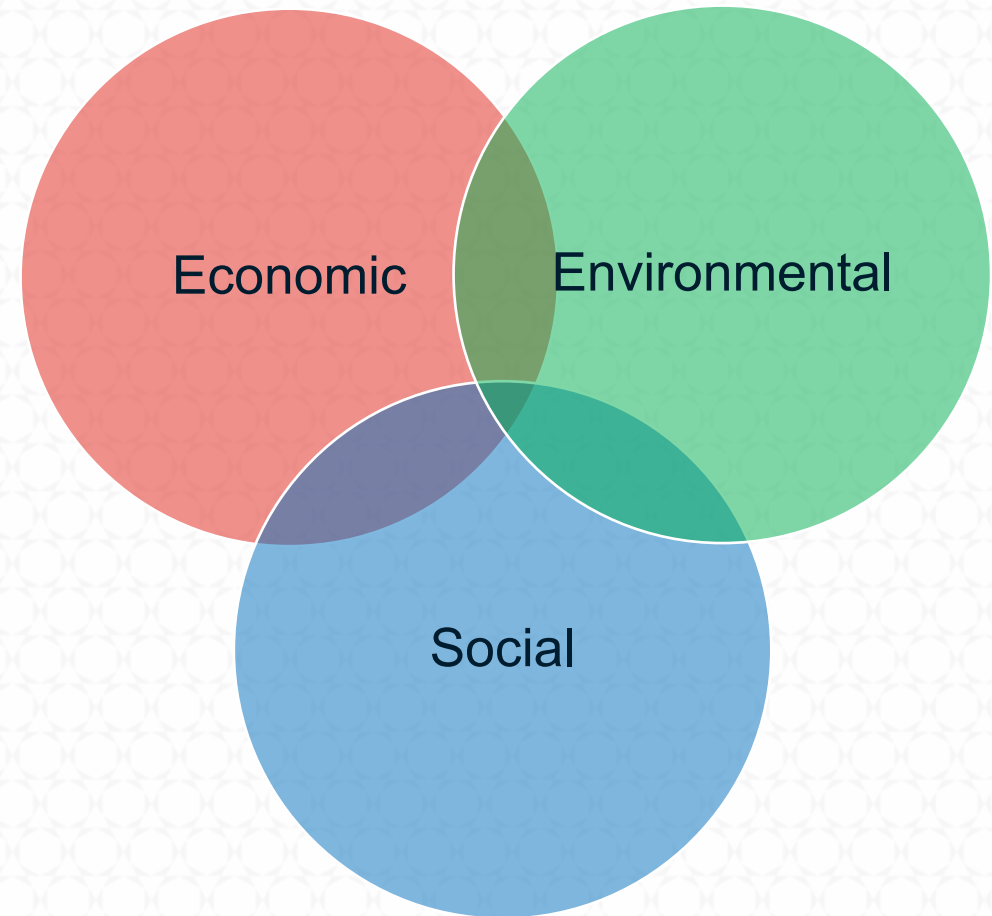
Uncertainty is the Source of Value



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The purpose of risk management is to create and protect value.

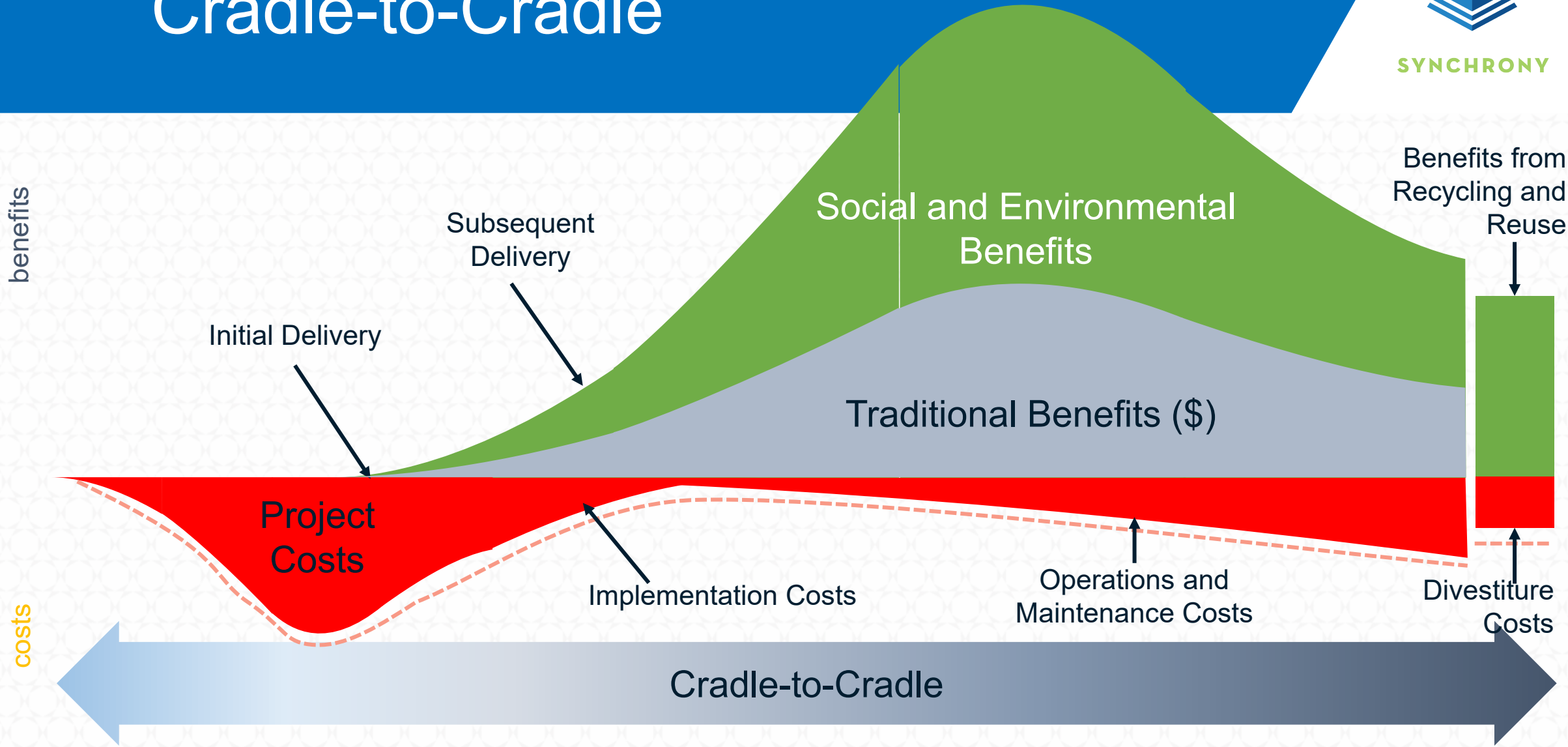
Value comes in different forms depending on what your organisation values.



Cradle-to-Cradle



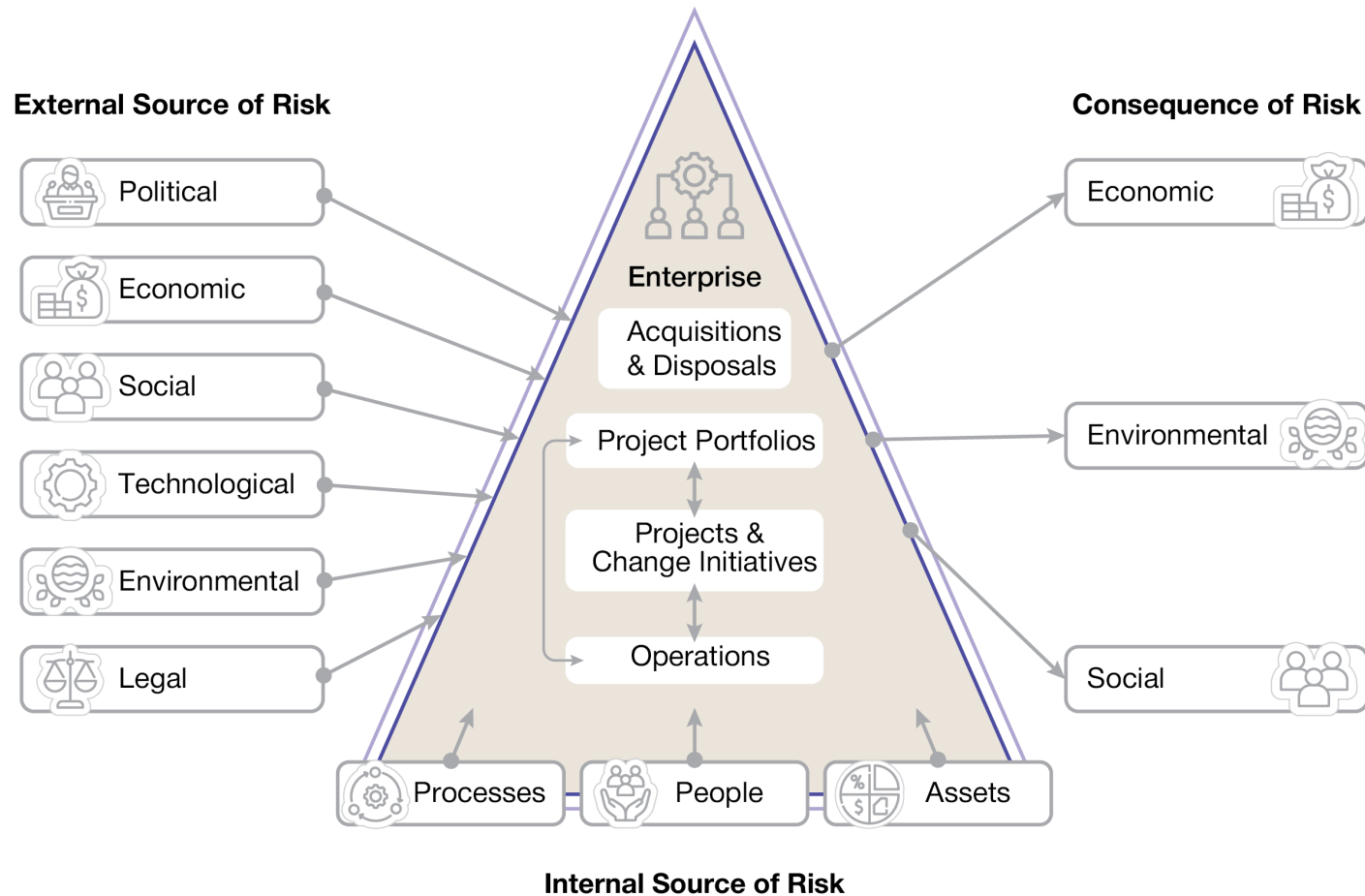
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“All Hazards” Risk and Value Assessment



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Perceptions



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**BLACK
AND
BLUE?**



**OR GOLD
AND
WHITE?**

Google Study – Project Aristotle



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What makes a team perform?

Psychological Safety

The belief that one can express oneself without fear of the consequences, or fear of being judged for speech, comment or action



Speaking Up without Fear



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When people feel safe to speak up,
projects avoid silent risks.

Ask yourself: Have you seen
someone afraid to raise an issue?



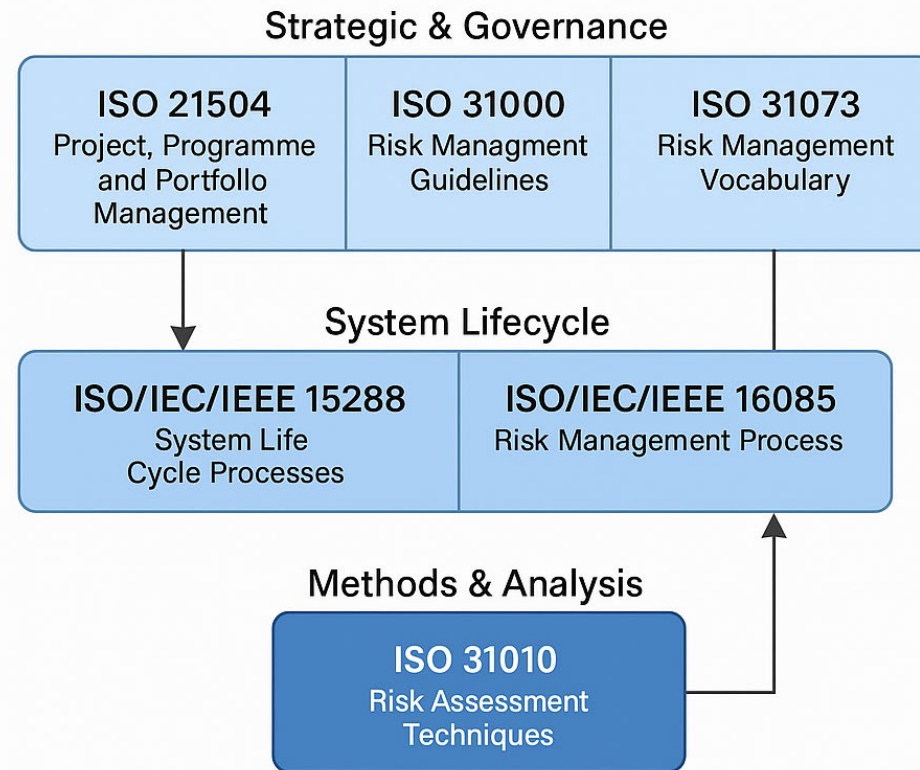
Governance and Standards

Reduce errors (noise and bias)



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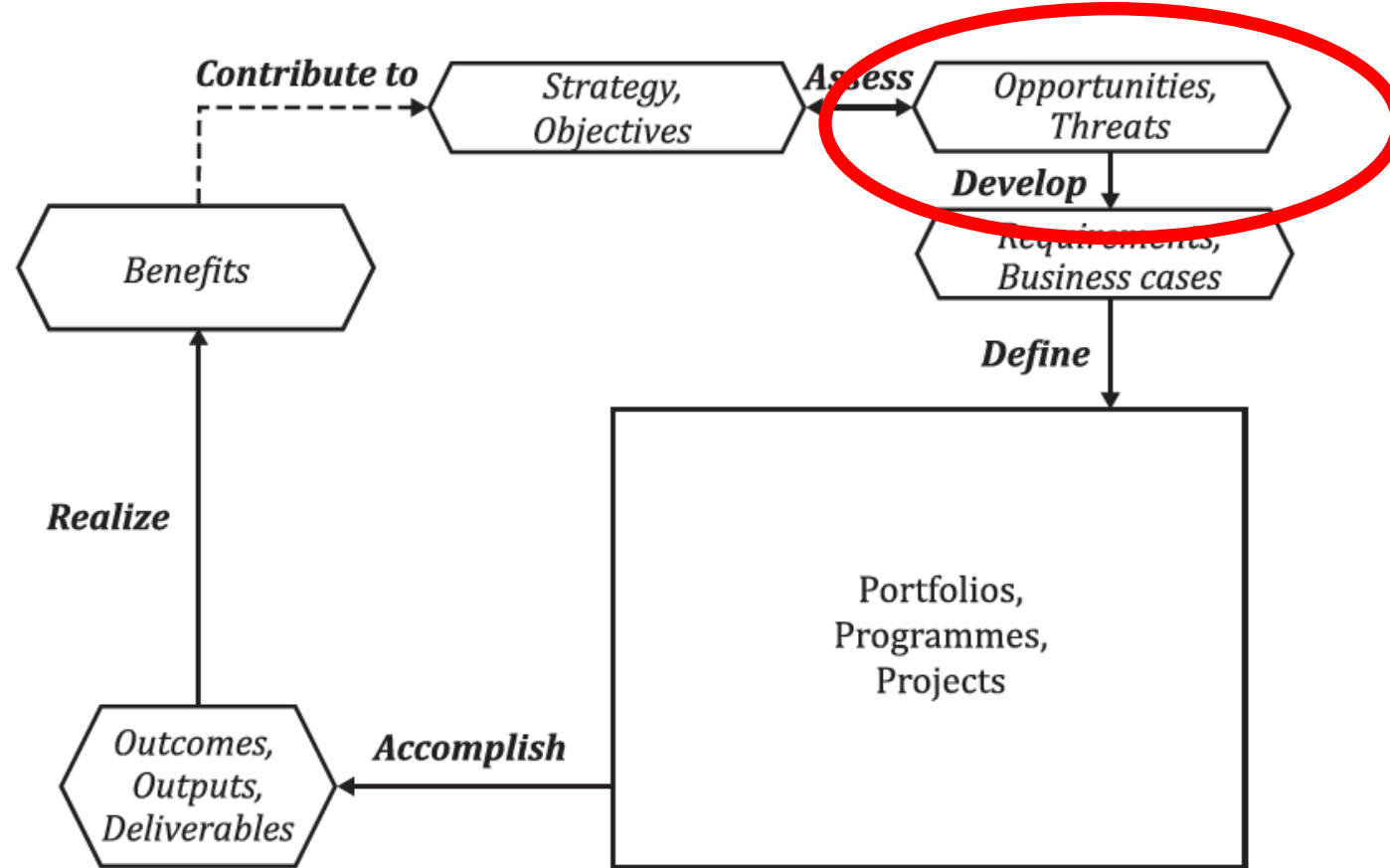
STANDARDS LINKING SYSTEMS THINKING AND RISK MANAGEMENT



How Organisations Deal with Uncertainty (ISO21504)



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NOTE Opportunities and threats are seen from a strategic perspective.

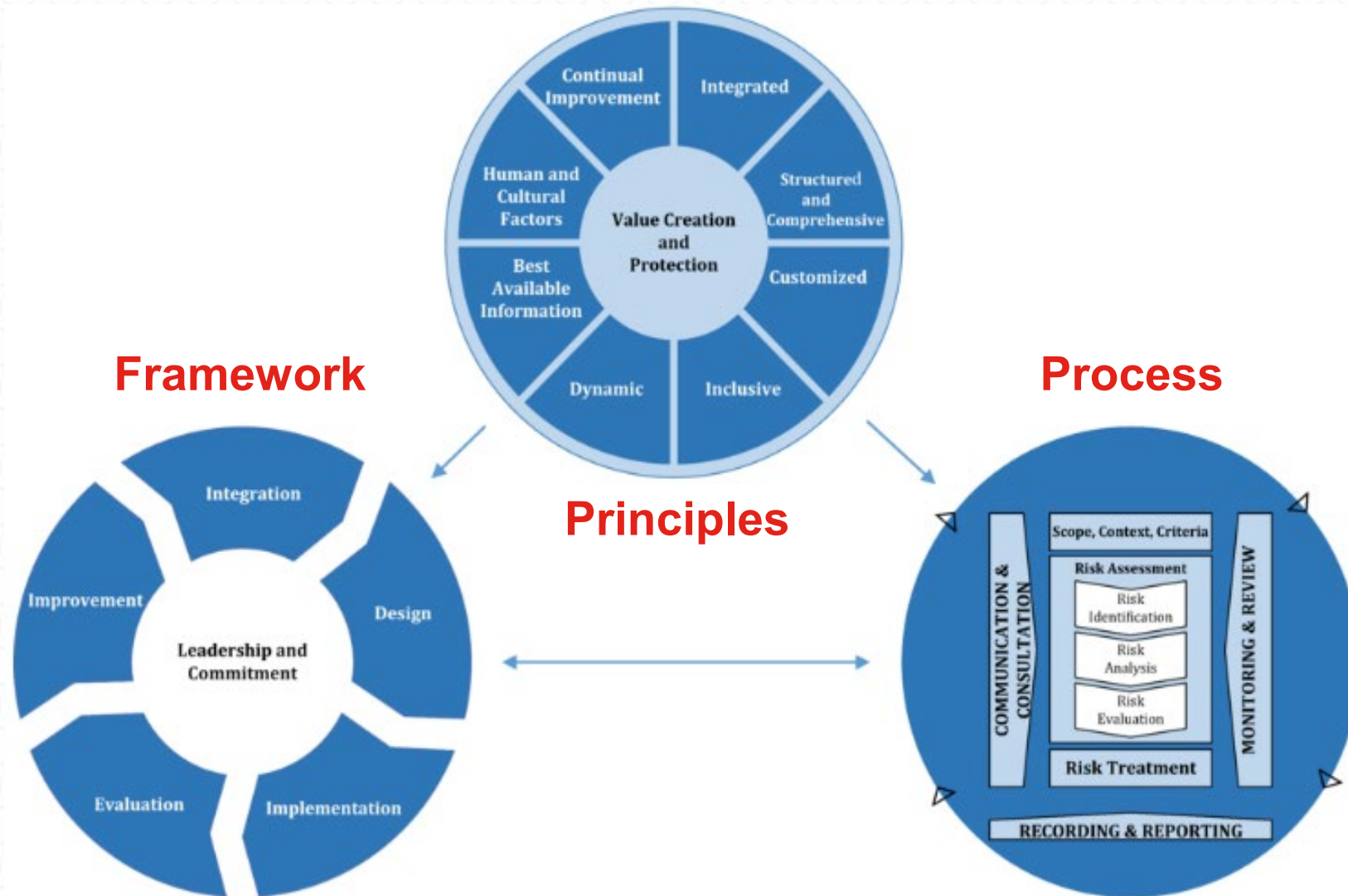
Figure 2 — An example of strategy implementation

ISO 31000

Guidelines for risk management



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ISO/IEC/IEEE 16085:2021, *System life cycle processes – Risk management*



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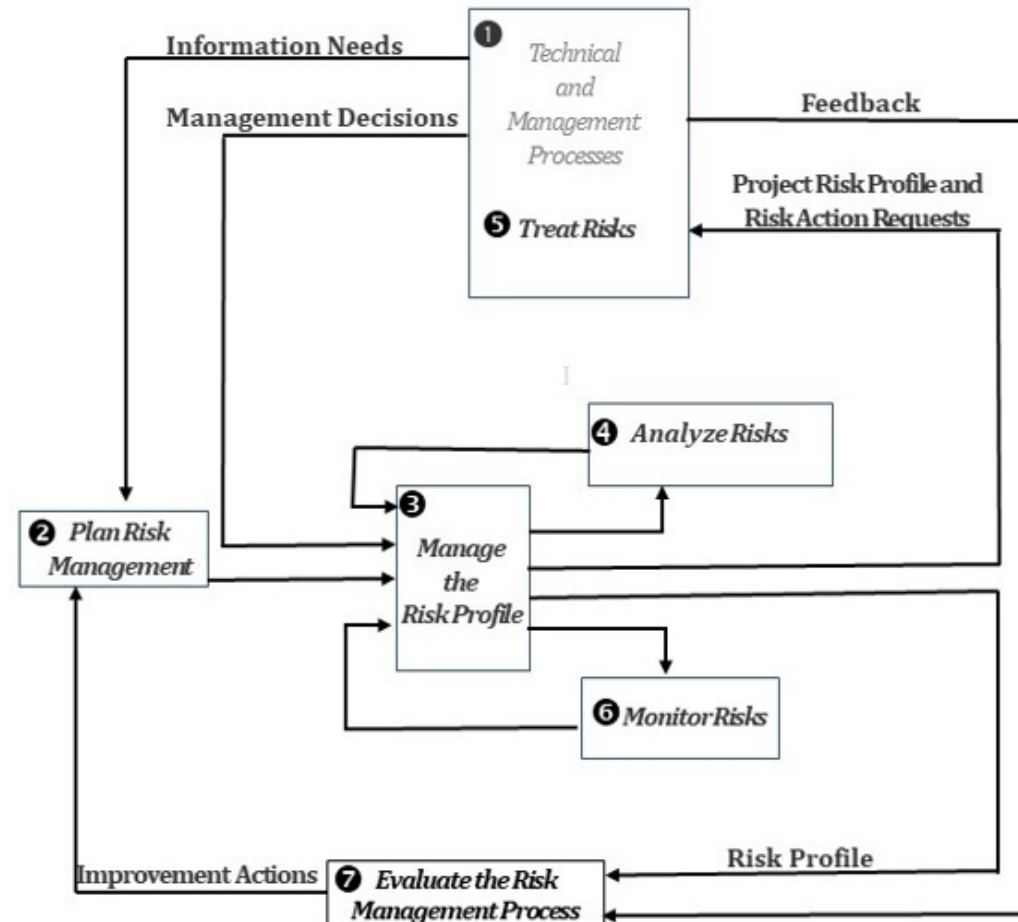
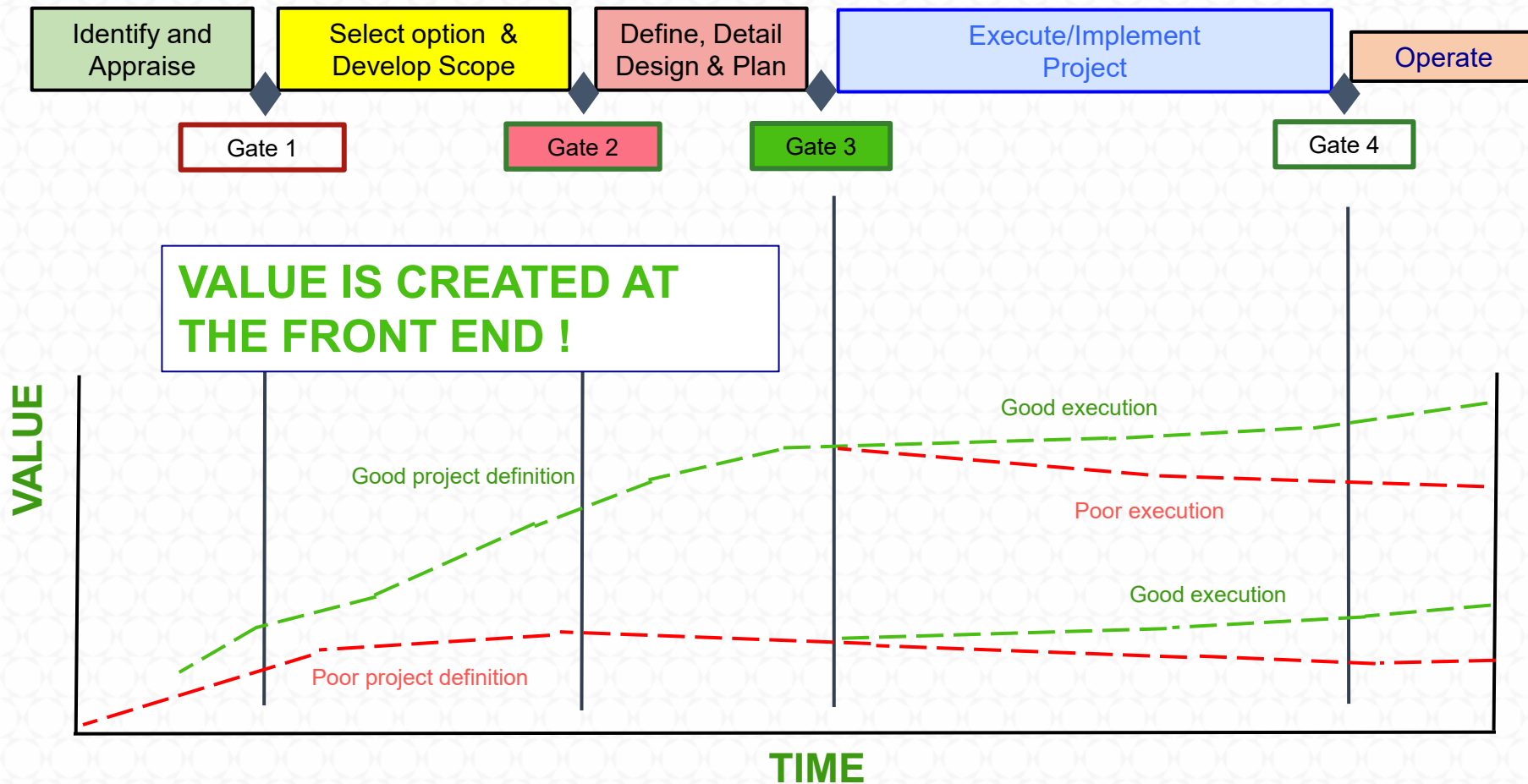


Figure 1 — Risk management process model

Value and Front-end Loading



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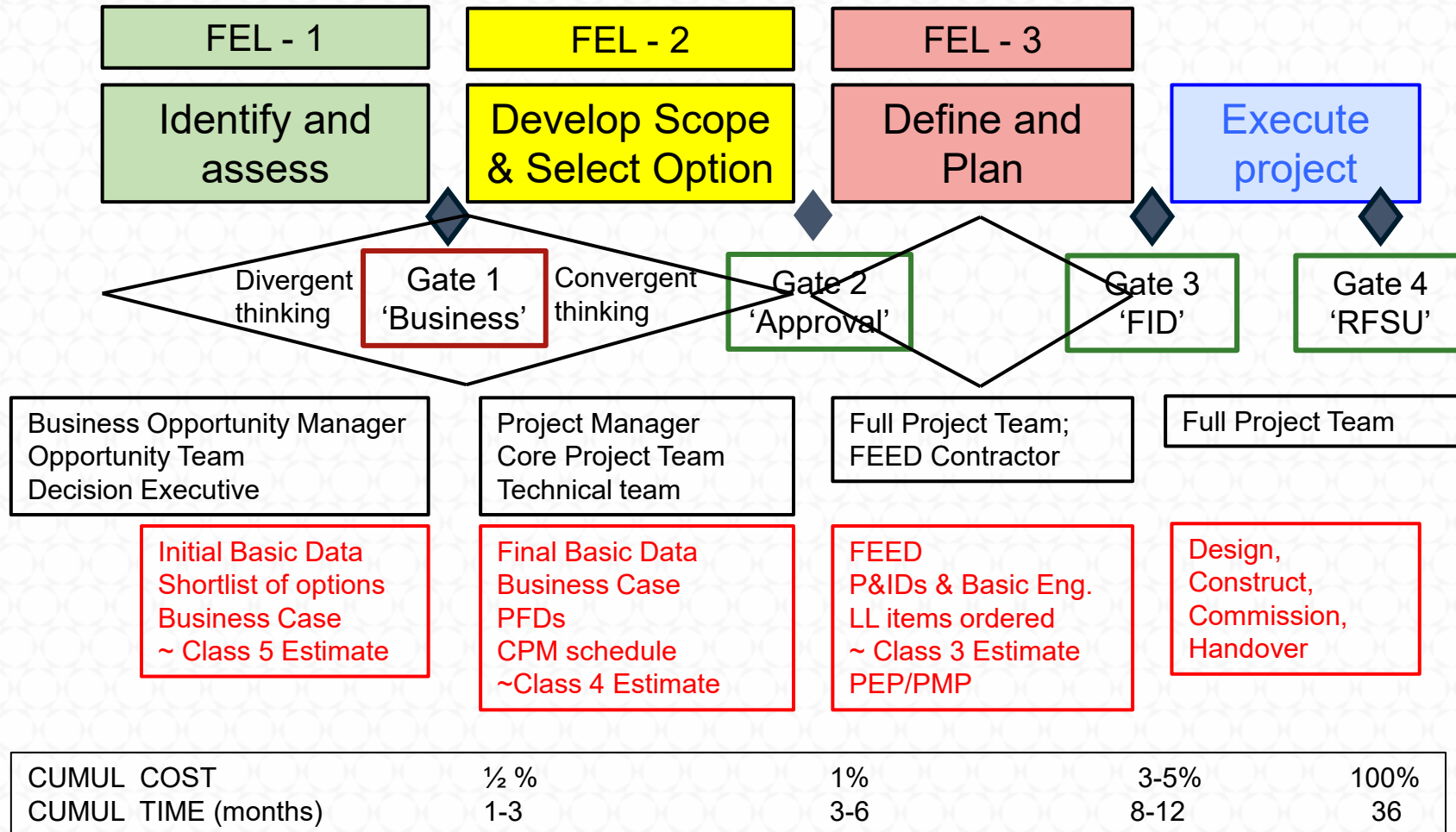


Front End Loading (FEL)

Divergent and Convergent Thinking



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Key take-aways



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-  **Awareness of bias improves decision quality.**
-  **Feedback Loops enable systemic learning**
-  **Open communication builds stronger teams.**
-  **Strong governance transforms uncertainty into opportunity**

Treat governance as a living system — one that learns, adapts, and creates value under uncertainty.

Thank you!

